

A2: TEC Heating and Cooling Analysis

Phys 39, Module 4 • Tianren Yin

1. Part 4: Combined graph and fits

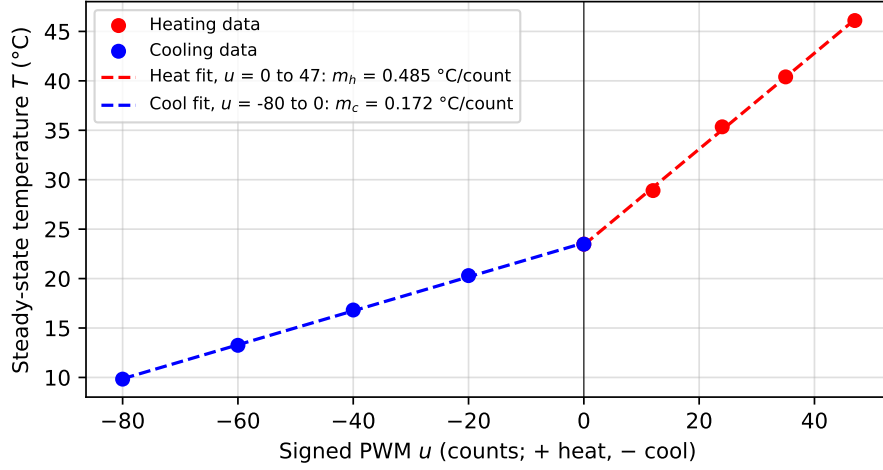


Figure 1: Steady-state temperature versus signed PWM u (positive = heating, red; negative = cooling, blue), with a least-squares line for each branch. Heating fit: $u = 0$ to 47; cooling fit: $u = -80$ to 0. Both branches share the $u = 0$ point (23.48°C). Steady state: after each PWM step we waited about three time constants ($\tau \lesssim 100\text{ s}$), then recorded the temperature once it stayed within $\pm 0.1^\circ\text{C}$ (about the normal measurement noise) for one further minute. Waits were 370–725 s.

2. Part 5.1: Measured slopes

Branch	Fit range (signed PWM)	Duty range D	Slope
Heating	0 to +47	0–0.184	$m_h = 0.485 \pm 0.007^\circ\text{C}/\text{count}$
Cooling	–80 to 0	0–0.314	$m_c = 0.172 \pm 0.002^\circ\text{C}/\text{count}$

$$r = \frac{m_h}{m_c} = \frac{0.485}{0.172} = 2.82 \pm 0.05.$$

(Uncertainties are the standard errors of the fitted slopes.) Neither branch shows visible curvature: the slopes between neighbouring points are $0.45, 0.54, 0.46, 0.48^\circ\text{C}/\text{count}$ (heating) and $0.17, 0.18, 0.17, 0.16^\circ\text{C}/\text{count}$ (cooling), with no systematic trend. We therefore fitted one straight line to all five points of each branch, from 0 to that direction’s maximum useful PWM.

3. Part 5.2: PWM and the slope-ratio model

Steady-state balance. At steady state $G(T - T_0) = \dot{Q}_{\text{TEC}}$. $\dot{Q}_{\text{TEC}}$ holds the two current-dependent object-face terms: Peltier transport (into the object when heating, out of it when cooling) and object-face Joule heating $\frac{1}{2}I^2 R_M$ (always into the object). $G(T - T_0)$ is the net passive heat flow out of the object, including conduction back through the TEC. When heating ($T > T_0$) it flows out; when cooling ($T < T_0$) it flows in. The individual flows are not zero; their sum is.

PWM averages. In one period τ the current is I for $D\tau$ and 0 for $(1 - D)\tau$, so

$$\langle I \rangle = \frac{1}{\tau} \left[\int_0^{D\tau} I dt + \int_{D\tau}^\tau 0 dt \right] = \frac{I D\tau}{\tau} = DI, \quad \langle I^2 \rangle = \frac{1}{\tau} \int_0^{D\tau} I^2 dt = DI^2.$$

Because the current is either full or zero, $\langle I^2 \rangle = DI^2 \neq \langle I \rangle^2 = D^2 I^2$ for $0 < D < 1$. Both terms are therefore linear in D , and the model predicts a straight line on each branch with a kink at $u = 0$. This matches our graph. With $\langle I^2 \rangle = D^2 I^2$ (as for DC), the heating branch would curve upward and the cooling branch would flatten. This assumes the PWM period ($\approx 2\text{ ms}$) is much shorter than the thermal time constant, so the block responds only to the averages.

Slope ratio. With signed duty $d = u/255$ and $D = |d|$, $\dot{Q}_{\text{TEC}} = d\dot{Q}_P + |d|\dot{Q}_J$.

$$\begin{aligned} \text{Heating } (d > 0, |d| = d) : \quad G(T_h - T_0) &= d(\dot{Q}_P + \dot{Q}_J) & \Rightarrow \quad \frac{dT_h}{dd} &= \frac{\dot{Q}_P + \dot{Q}_J}{G}, \\ \text{Cooling } (d < 0, |d| = -d) : \quad G(T_c - T_0) &= d(\dot{Q}_P - \dot{Q}_J) & \Rightarrow \quad \frac{dT_c}{dd} &= \frac{\dot{Q}_P - \dot{Q}_J}{G}. \end{aligned}$$

Both slopes are positive when $\dot{Q}_P > \dot{Q}_J$, and the heating slope is larger. Since $m = dT/du = \frac{1}{255} dT/dd$, the factors $1/255$ and G cancel:

$$r = \frac{m_h}{m_c} = \frac{\dot{Q}_P + \dot{Q}_J}{\dot{Q}_P - \dot{Q}_J}.$$

Let $x = \dot{Q}_J/\dot{Q}_P$. Then $r = (1+x)/(1-x)$, so $r - rx = 1 + x$, $r - 1 = x(r+1)$, and

$$\boxed{\frac{\dot{Q}_J}{\dot{Q}_P} = \frac{r-1}{r+1}} = \frac{2.82-1}{2.82+1} = 0.48 \pm 0.01.$$

(Check: $r = 2$ gives $1/3$.) At our on-state current, the object-face Joule heat is about half of the Peltier heat pumping.

4. Part 5.3: Laird data-sheet calculation

Source: Laird CP14-127-045-L2-W4.5 data sheet (Rev. 00, 06-01-2022), p.3, *Specifications* table, hot-side temperature $T_h = 27.0^\circ\text{C}$ column.

Quantity	Value	Meaning and operating condition ($T_h = 27^\circ\text{C}$)
R_M	1.50 Ω	Electrical resistance of the module.
$I_{\max}$	8.6 A	Current that produces $\Delta T_{\max}$ with no heat load ($Q_c = 0$).
$Q_{c,\max}$	71.3 W	Largest heat pumped from the cold side, at $I = I_{\max}$ and $\Delta T = 0$.
$\Delta T_{\max}$	70.5 $^\circ\text{C}$	Largest hot-to-cold temperature difference, at $I = I_{\max}$ and $Q_c = 0$.

At $\Delta T = 0$ the conduction term is zero. Half of the Joule heat goes to the object face:

$$\dot{Q}_{J,\max} = \frac{1}{2} I_{\max}^2 R_M = \frac{1}{2} (8.6 \text{ A})^2 (1.50 \Omega) = 55.5 \text{ W}, \quad \dot{Q}_{P,\max} = Q_{c,\max} + \dot{Q}_{J,\max} = 71.3 + 55.5 = 126.8 \text{ W},$$

$$r_{\text{Laird},\max} = \frac{\dot{Q}_{P,\max} + \dot{Q}_{J,\max}}{\dot{Q}_{P,\max} - \dot{Q}_{J,\max}} = \frac{126.8 + 55.5}{126.8 - 55.5} = \frac{182.3}{71.3} = 2.56.$$

At $I_{\max}$ this corresponds to $\dot{Q}_J/\dot{Q}_P = 55.5/126.8 = 0.44$.

5. Part 5.4: Compare the ratios

Measured: $r = 2.82$ ($\dot{Q}_J/\dot{Q}_P = 0.48$). Laird at $I_{\max}$: $r = 2.56$ (0.44). The measured ratio is about 10% higher, which is similar but not equal. Exact agreement is not expected:

- **$D = 1$ does not mean $I = I_{\max}$.** Full duty only keeps the H-bridge on. The on-state current is set by the 12 V supply, the H-bridge and wiring drops, R_M , and the TEC's Seebeck back-voltage. That gives at most $12 \text{ V}/1.50 \Omega \approx 8 \text{ A}$ and in practice less, below $I_{\max} = 8.6 \text{ A}$. The 10 A supply limit is not reached.
- **The Joule fraction scales with current:** $\dot{Q}_J/\dot{Q}_P = \frac{1}{2} I^2 R_M / (S_M T_o I) \propto I$. A lower current alone would make r *smaller* than 2.56, so the excess comes from the effects below.
- **Peltier term $\propto T_o$ (kelvin).** The object face averages $\approx 308 \text{ K}$ on the heating branch and $\approx 290 \text{ K}$ on the cooling branch, so Peltier pumping is relatively stronger when heating. This raises r .
- **Finite ΔT , PWM rather than DC, changing properties.** The data sheet is for steady DC at $\Delta T = 0$. Our object face spans $10\text{--}46^\circ\text{C}$, which changes TEC conduction and the back-voltage. R_M itself rises from 1.50 to 1.68 Ω between 27 and 50°C (same table), and S_M and K_M also vary.
- **Passive paths and fitting.** If the leaks are not equal above and below room temperature, G does not cancel exactly. One straight line per branch also averages over any slight curvature.

6. Part 5.4: Passive conduction

When the object is hotter than its surroundings (heating), passive heat flows *out of* the object; when it is colder (cooling), passive heat flows *into* it. Either way it pulls the temperature back toward T_0 , so it opposes both heating and cooling. Approximately symmetric conduction enters both branches through the same G . It reduces both slopes by the same factor $1/G$, which cancels from r . Without Joule heating the two branches would be $\pm \dot{Q}_P/G$, equal in magnitude. The unequal slopes come from Joule heating, which keeps its sign when the current reverses: it adds to Peltier heating and subtracts from Peltier cooling.